Descrete Math ch 11: Eq. Relation:

Equivalence Relation: Let S be a set and R a relation on S. We say that R is an equivalence relation if R is reflexive, symmetric, and transitive.

For all $x,y,z \in S$, if $(x,y) \in R$ and $(y,z) \in R$, then $(x,z) \in R$
"if x is related to y and y is related to z then x is related to z"

for every $x \in S$, we have $(x,x) \in R$
"Everything is related to itself"

for every $x,y \in S$ if $(x,y) \in R$, then $(y,x) \in R$.
" if x is related to y then y is related to x"

Modular Equivalence: let $n \in \mathbb{N}$, we define a relation on $\mathbb{Z}$:

$$(a,b) \in \equiv \text{ if and only if } n \mid (a-b)$$

and we write it as $a \equiv b \pmod n$

"read as a is congruent to b modulo n"

Ex) $n=5$

$12 \equiv 2 \pmod 5$ because $5 \mid (12-2)$

$17 \equiv 2 \pmod 5$ " $5 \mid (17-2)$

So all $12, 17, 2, -3, 8, \ldots$ are equivalent modulo 5 (they all give remainder 2 when devided by 5)

★ obv. this is a relation but we need to show this is equivalence relation

Let $n \in \mathbb{N}$ and we let $\equiv$ be the relation "congruent modulo n" on $\mathbb{Z}$
then $\equiv$ is an equivalence relation.

To Prove this state we need to show: ① Reflexive ② Symmetric ③ transitive.

Reflexive: we must show that: for any integer x, we have $x \equiv x \pmod n$

$x - x = 0$

$n \cdot 0 = 0$, so n devides 0

$n \mid (\underbrace{x-x}_{0})$

Therefore $(x,x) \in \equiv$.

Symmetric: we must show: if $x \equiv y \pmod n$, then $y \equiv x \pmod n$

Assume $(x, y) \in \equiv$. this means:

$n \mid (x-y) \longrightarrow$ so there is some integer t that

$$nt = x - y$$

now multiply both sides by -1:

$$n(-t) = y - x$$

Since $-t$ is still an integer so $n \mid (y-x)$ so $(y, x) \in \equiv$.

transitive: if we have $x \equiv y \pmod n$ and $y \equiv z \pmod n$, then $x \equiv z \pmod n$

Assume:

$$\left.\begin{array}{l} n \mid (x-y) \longrightarrow ns = x - y \\ n \mid (y-z) \longrightarrow nt = y - z \end{array}\right\} \begin{array}{l} ns + nt = x - y + y - z \\ ns + nt = x - z \\ n(s+t) = x - z \end{array}$$

Since $s+t$ is an integer so $n \mid (x-z)$, Thus $(x, z) \in \equiv$

11.1. *Define relation* R *on* $\mathbb{Q}$ *via* $R = \{(a, b) : a - b \in \mathbb{Z}\}$. *Prove that* R *is an equivalence relation.* [Hint 35]

this is basically saying two rationals are related if their difference is an integer.

we must show R is reflexive, symmetric, and transitive:

Reflexive Part:

take $a \in \mathbb{Q}$

$a - a = 0$

and $0 \in \mathbb{Z}$

So $a - a \in \mathbb{Z}$, hence $(a, a) \in R$, thus R is reflexive.

Symmetric Part:

Assume $(a, b) \in R$

this means that $a - b \in \mathbb{Z}$

But then

$b - a = -(a - b)$

the integer are closed under negation. so if $(a-b) \in \mathbb{Z}$ then $-(a-b) \in \mathbb{Z}$

so $b - a \in \mathbb{Z}$ which means $(b, a) \in R$

Transitive Part:

Assume $(a, b) \in R$ and $(b, c) \in R$

so $a - b \in \mathbb{Z}$, $b - c \in \mathbb{Z}$

let $m = a - b$ and $n = b - c$, where $m, n \in \mathbb{Z}$

then

$a - c = (a - b) + (b - c) = m + n$

Since the sum of two integers is an integer, $m + n \in \mathbb{Z}$.

So $a - c \in \mathbb{Z}$, So $a - c \in R$.

11.2. *Define relation* R *on* $\mathbb{Q}$ *via* $R = \{(a, b) : a + b \in \mathbb{Z}\}$. *Prove that* R *is not an equivalence relation.* [Hint 18]

We need to show that at least one of equivalence relation condition fails.

reflexive:

$\forall a \in \mathbb{Q}, (a, a) \in R$

that is $\forall a \in \mathbb{Q}$, $a + a = 2a \in \mathbb{Z}$

we use counter example. let $a = \frac{1}{4}$

than $2a = \frac{1}{2} \notin \mathbb{Z}$

So $(\frac{1}{4}, \frac{1}{4}) \in R$

For exercises 11.3-11.5, $\mathbb{Z}[x]$ is the set of all polynomials with integer coefficients, in the variable x. Six examples of elements of $\mathbb{Z}[x]$ are:

$5x^2 - 2x + 1, \; -x^3 - 7, \; 100x^{100} + x^{99}, \; x, \; 8, \; 0$

If $p(x) \in \mathbb{Z}[x]$, it is a polynomial, and you can do anything to it that you do with other polynomials. For example, if $p(x) = -x^3 - 7$, we can calculate $p(0) = -0^3 - 7 = -7$, $p(x) + p(x) = -2x^3 - 14$, $p'(x) = -3x^2$.

11.3. *Define relation* R *on* $\mathbb{Z}[x]$ *via* $R = \{(p(x), q(x)) : p(0) - q(0) = 0\}$. *Prove that* R *is an equivalence relation.* [Hint 187]

We need to show two polynomials are related iff they have the same value at zero.

Reflexive: take any polonymial $p(x)$

$P(0) - P(0) = 0$, so $(P(x), P(x)) \in R$

Symmetric: Assume $(P(x), q(x)) \in R$

then $P(0) - q(0) = 0 \Rightarrow P(0) = q(0)$

Then $q(0) - P(0) = 0$

So $(q(x), P(x)) \in R$

Transitative: Asume $(P(x), (q(x)) \in \mathbb{Z}$ and $(q(x), r(x)) \in R$

So $p(0) = q(0)$ and $q(0) = r(0)$ so $P(0) = r(0)$

So $P(0) - r(0) = 0$

So $((P(x), r(x)) \in R$

11.4. *Define relation* R *on* $\mathbb{Z}[x]$ *via* $R = \{(p(x), q(x)) : p(0) + q(0) = 0\}$. *Prove that* R *is not an equivalence relation.* [Hint 82]

Reflexive?

we use counterexample to show this is not reflexive:

take $P(0) = 1$ then $P(1) + P(1) = 2 \neq 0$ Thus $(P(x), P(x)) \notin R$

$P(x), q(x) \in R$ $q(x), t(x) \in R$

11.5. *Define relation* R *on* $\mathbb{Z}[x]$ *via* $R = \{(p(x), q(x)) : p'(x) = q'(x)\}$. *Here* $'$ *denotes the derivative operator:* $p'(x)$ *is the derivative of* $p(x)$, *and* $q'(x)$ *is the derivative of* $q(x)$. *Prove that* R *is an equivalence relation.* [Hint 281]

Reflexive:

take any Polynomial $P(x)$

of course: $P'(x) = P'(x)$ for all x, so $(P(x), P(x)) \in R$

Symmetric:

Assume $((P(x), q(x)) \in R$, that means $P'(x) = q'(x)$

But the equality is symmetric, so also

$q'(x) = p(x)$

Hence $(q(x), P(x)) \in R$

Transitive:

Assume $(P(x), q(x)) \in R$ and $(q(x), r(x)) \in R$

so this means: $P'(x) = q'(x)$ and $q'(x) = r'(x)$

by transitivity of equality we have

$P'(x) = r'(x)$

Thus, $(P(x), r(x)) \in R$.

11.6. *The Euclidean Algorithm Theorem (proved in a more advanced course) says that for any* $x, y \in \mathbb{Z}$ *(not both zero), there must exist* $a, b \in \mathbb{Z}$ *satisfying* $ax + by = \gcd(x, y)$.
Now, fix $n \in \mathbb{N}$*, and consider the relation* $\equiv_n$ *on* $\mathbb{Z}$*. Prove that for any* $m \in \mathbb{Z}$ *there is some* $m' \in \mathbb{Z}$ *so that* $mm' \equiv_n \gcd(m, n)$. *[Hint 294]*

we fix $n \in \mathbb{N}$ and use the fact that for any integers x, y (not both 0) there exist $a, b \in \mathbb{Z}$ with $ax + by = \gcd(x, y)$

we take $x = m, y = n$ ★

so there exist some integers a, b such that

$am + bn = \gcd(m, n)$

Now we want some m' with $mm' \equiv_n \gcd(m, n)$ which means $n \mid (mm' - \gcd(m, n))$

Just choose $m' = a$ from ★

So $mm' - \gcd(m, n) = ma - (am + bn) = -bn = (-b)n$

Since $-b$ is multiple of n, we have

$n \mid mm' - \gcd(m, n))$, $mm' \equiv_n \gcd(m, n)$.

Thus for every integer m there exist an integer m' (namely $m' = a$ from the Euclidean algorithm representation) such that

$mm' \equiv_n \gcd(m, n)$.

Fix $n \in \mathbb{N}$

Assume

- $x_1 \equiv y_1 \pmod{n}$
- $x_2 \equiv y_2 \pmod{n}$

Then:

1. $x_1 + x_2 \equiv y_1 + y_2 \pmod{n}$
2. $x_1 - x_2 \equiv y_1 - y_2 \pmod{n}$
3. $x_1 y_1 \equiv x_2 y_2 \pmod{n}$

* So if two numbers are congruent, you can replace them by each other in sums, differences and products.

This is helpful because we can do stuff like:

- if $x \equiv 5 \pmod{11}$, then

$2x \equiv 2.5 \pmod{11}$

$x^2 \equiv 5^2 = 25 \equiv 3 \pmod{11}$, etc

Modular division (the "cancel" rule):

Let $m, n \in \mathbb{N}$ and $x, y \in \mathbb{Z}$

The following are equivalent.

- "$mx \equiv my \pmod{n}$"
- "$x \equiv y \pmod{\frac{n}{\gcd(m,n)}}$"

- So you cannot always just cancel m from both sides of $mx \equiv my \pmod{n}$
- But you can cancel if you also adjust the modulos from n to $n/\gcd(m,n)$

$$mx \equiv my \pmod{n} \longleftrightarrow x \equiv y \pmod{n/\gcd(m,n)}$$

Collary (Nice cases): $\gcd(m,n) = 1$

if $\gcd(m,n) = 1$ (i.e. m and n are coprimed), Then $n/\gcd(m,n) = n$

So in this case we have $mx \equiv my \pmod{n} \longleftrightarrow x \equiv y \pmod{n}$

So you can cancel m without chaning the modulus

Collary (case $m|n$): if m devides n, then $\gcd(m,n) = m$ and then $n/\gcd(m,n) = n/m$

So the theorem becomes $mx \equiv my \pmod{n} \longleftrightarrow x \equiv y \pmod{n/m}$.

Example: solving $3x \equiv 6$.

Mod 11: Eq. $3x \equiv 6 \pmod{11}$

Here $\gcd(3,11) = 1$

$x \equiv 2 \pmod{11}$

All solutions: $\dots, -9, 2, 13, 24, \dots$

between $0, \dots, 10$ we have

$x = 2$

Mod 12: Eq $3x \equiv 6 \pmod{12}$

$3 \mid 12$ so we can divide both sides and the modulus by 3:

$x \equiv 2 \pmod 4$

All solutions: $\dots, -6, -2, 2, 6, 10, 14, 18, \dots$

if we look at $0, \dots, 11$ we have $2, 6, 10$

Chinese Remainder theorem: (two moduli):

if $m, n \in \mathbb{N}$ with $\gcd(m,n) = 1$, and integers a, b, then the system

$$\begin{cases} x \equiv a \pmod m \\ x \equiv b \pmod n \end{cases}$$

has exactly 1 solution in the range $0 \leq x < mn$.

So if moduli are coprime, we can solve 2 modular equation at once, and there will be a unique solution modulo mn.

Proof add/subtract/multiply $\quad$ $u \equiv v \pmod n$ means $n \mid (u - v)$

Add: Let $n \in \mathbb{N}$ and suppose $x_1 \equiv y_1 \pmod n$, $x_2 \equiv y_2 \pmod n$

$n \mid (x_1 - y_1) \qquad n \mid (x_2 - y_2)$

$(x_1 - y_1) - (x_2 - y_2) = (x_1 + x_2) + (y_1 + y_2)$

So $n \mid (x_1 + x_2 - (y_1 + y_2))$

$x_1 + x_2 \equiv y_1 + y_2 \pmod n$

Multiply) $x_1 = y_1 + kn \qquad x_2 = y_2 + ln \qquad$ for some integers k, l.

$x_1 x_2 = (y_1 + kn)(y_2 + ln)$

$x_1 x_2 = y_1 y_2 + y_1 ln + y_2 kn + kln^2$

$x_1 x_2 = y_1 y_2 + n(y_1 l + y_2 k + kln)$

So $x_1 x_2 - y_1 y_2 = n(y_1 l + y_2 k + kln)$

$x_1 x_2 \equiv y_1 y_2 \pmod n$

subtract) $(x_1 - x_2) - (y_1 - y_2) = (x_1 - y_1) - (x_2 - y_2)$

$n \mid (x_1 - y_1) - (x_2 - y_2)$

$x_1 - y_1 \equiv x_2 - y_2 \pmod n$

Proof Modular Division:

Let $m, n \in \mathbb{N}$ and $x, y \in \mathbb{Z}$, let $d = \gcd(m, n)$ then the following equivalent: a. $mx \equiv my \pmod n$ b. $x \equiv y \pmod{\frac{n}{d}}$

set: $d = \gcd(m, n)$

$m = dm'$, $n = dn'$ with $\gcd(m', n') = 1$

① Direction a. $\rightarrow$ b:

Assume $mx \equiv my \pmod n$ that means $n \mid mx - my$

$n \mid m(x-y) = dn' \mid dm'(x-y) = n' \mid m'(x-y)$ given the fact that m' and n' share no factors, all factors that make n' devide the product comes from $x - y$.

So we get $n' \mid (x-y) \longleftrightarrow x \equiv y \pmod{n'}$ but also

$n' = \frac{n}{d}$ so $x \equiv y \pmod{n/d}$, which is b

② Direction b. $\rightarrow$ a.

Now assume $x \equiv y \pmod{n/d}$ which means $\frac{n}{d} \mid (x-y)$

So there exist an integer k such that $x - y = k\frac{n}{d}$, multiply both sides by $m = dm'$:

$$m(x-y) = dm'(x-y) = dm'k\frac{n}{d} = m'kn$$

So n devides $m(x-y)$, $n \mid m(x-y)$ that is $mx \equiv my \pmod n$ which is a.

So $a \longleftrightarrow b$ proved.

Example for CRT: Solve $\begin{cases} x \equiv 2 \pmod 3 \\ x \equiv 3 \pmod 5 \end{cases}$

Solution:

So 3, 5 are coprime therefore there exists exactly one solution between $[0, 15)$.

$x \equiv 2 \pmod 3$

2, 5, 8, 11, 14, ... $\rightarrow 3k + 2$, we need one of this also satisfy $x \equiv 3 \pmod 5$

2 mod 5 = 2 ✗ 5 mod 5 = 0 ✗ 8 mod 5 = 3 ✓ so $x = 8$ works

11.7. *Let* $n \in \mathbb{N}$, *and define relation* R *on* $\mathbb{Z}$ *via* $(a, b) \in R$ *if* a, b *have the same remainder upon dividing by* n *(via the Division Algorithm). Prove that* $(a, b) \in R$, *if and only if,* $a \equiv b \pmod{n}$. *This gives another way of understanding modular equivalence.* [Hint 283]

we need to prove two directions:

(1) same reminder $\Rightarrow a \equiv b \pmod{n}$

(2) $a \equiv b \pmod{n} \Rightarrow$ same remainder

we use devision alghorithm: $(1) \rightarrow (2)$:

Assume $(a,b) \in R$, a,b have same remainder when devided by n

so there exist integers p,q and remaider r with:

$$a = qn + r, \quad b = pn + r, \quad 0 \leq r < n$$

subtract:

$$a - b = (qn + r) - (pn + r) = (q - p)n$$

$$n \mid (a - b) \longleftrightarrow a \equiv b \pmod{n}$$

$(2) \rightarrow (1)$:

Now assume $a \equiv b \pmod{n}$ that means $n \mid (a - b)$ i.e. $a - b = kn$ for some $k \in \mathbb{Z}$.

write a,b using devision alghoritm with their remainders:

$$a = qn + r \quad 0 \leq r < n$$

$$b = pn + s \quad 0 \leq s < n$$

for some integers q, p, r, s.

subtract $a - b = (qn + r) - (pn + s) = (q - p)n + (r - s)$

$$kn = (q - p)n + (r - s)$$

$$(r - s) = (q - p)n - kn = (k - (q - p))n$$

so $(r - s)$ is multiple of n.

but r and s are both between 0 and $n-1$

so $r - s$ is between $-(n-1)$ and $n-1$

the only interval of n in that interval is 0.

so $r - s = 0$, $r = s$

so a,b have same remainder when devided by n. that is $(a,b) \in R$

11.8. *Let* $m, n \in \mathbb{N}$. *Suppose that* $x \equiv y \pmod{mn}$. *Prove that* $x \equiv y \pmod{m}$, *and that* $x \equiv y \pmod{n}$. *[Hint 180]*

$x \equiv y \pmod{mn}$ that means $mn \mid (x-y)$, so there exists some integers k with $x-y = kmn$

(1) show $x \equiv y \pmod{m}$

Since $x-y = kmn$, we can factor $x-y = (kn)m$. So $x-y$ is multiple of m so $m \mid x-y \to x \equiv y \pmod{m}$

(2) since $x-y = kmn = (km)n$, so $n \mid (x-y) \to x \equiv y \pmod{n}$

11.10. *Find an integer* $x \in [0, 11)$ *such that* $x \equiv 2^8 \pmod{11}$. *Then find an integer* $y \in [0, 11)$ *such that* $y \equiv x \cdot x \pmod{11}$. *Note that* $y \equiv 2^{16} \pmod{11}$. *[Hint 42]*

we want $x \in [0, 11]$ with $x \equiv 2^8 \pmod{11}$, then $y \in [0, 11]$ with $y \equiv x \cdot x \pmod{11}$.

$2^1 = 2 \equiv 2 \pmod{11}$

$2^2 = 4 \equiv 4$ "

$2^3 = 8 \equiv 8$ "

$2^4 = 16 \equiv 5$ "

$2^5 = 32 \equiv 10$ "

$2^6 \equiv 9$ "

$2^7 \equiv 7$ "

$2^8 \equiv 3$ "

$y \equiv x \cdot x \pmod{11}$

$y \equiv 9 \pmod{11}$

$y = 9$ and since $2^8 = 3$

$2^{16} = (2^8)^2 \equiv 3^2 = 9 \equiv y \pmod{11}$

Answer $x = 3$

$y = 9$

11.11. *Use the method of Exercise 11.10 to find an integer* $z \in [0, 11)$ *such that* $z \equiv 2^{128} \pmod{11}$. *Do not actually compute* 2^{128}. *[Hint 268]*

$2^8 \equiv 3 \pmod{11}$

$2^{16} = (2^8)^2 \equiv 3^2 = 9 \pmod{11}$ So $y = 9$

$2^{16} \equiv 9 \pmod{11}$ $\quad 2^{32} = (2^{16})^2 \equiv 9^2 = 81 \equiv 4 \pmod{}$

$2^{64} = (2^{32})^2 \equiv 4^2 = 16 \equiv 5 \pmod{11}$

$2^{128} = (2^{64})^2 \equiv 5^2 = 25 \equiv 3 \pmod{11}$ So $\boxed{z = 3}$

11.12. *Use the work from Exercise 11.11, and the observation that* $100 = 64 + 32 + 4$*, to find an integer* $z \in [0, 11)$ *such that* $z \equiv 2^{100} \pmod{11}$*. Do not actually compute* 2^{100}*.* *[Hint 147]*

$2^{100} = 2^{64} \cdot 2^{32} \cdot 2^{4}$

$2^{64} \equiv 5,\ 2^{32} \equiv 4,\ 2^{4} \equiv 5 \pmod{11}$

$2^{100} \equiv 2^{64} \cdot 2^{32} \cdot 2^{4} \pmod{11}$

$2^{100} \equiv 5 \cdot 4 \cdot 5 \pmod{11}$

$5 \cdot 4 = 20 \equiv 9 \pmod{11}$

$9 \cdot 5 = 45 \equiv 45 - 44 = 1 \pmod{11}$

so $2^{100} \equiv 1 \pmod{11}$ so $\boxed{z = 1}$

11.13. *Compute* $2^1 \pmod{11}, 2^2 \pmod{11}, 2^3 \pmod{11}, \ldots,$ *and find a pattern. Use that pattern to find an integer* $z \in [0, 11)$ *such that* $z \equiv 2^{100} \pmod{11}$*.* *[Hint 70]*

$2^1 = 2 \equiv 2 \pmod{11}$
$2^2 = 4 \equiv 4$ "
$2^3 = 8 \equiv 8$ "
$2^4 = 16 \equiv 5$ "
$2^5 = 32 \equiv 10$ "
$2^6 \equiv 9$ "
$2^7 \equiv 7$ "
$2^8 \equiv 3$ "
$2^9 \equiv 6$ "
$2^{10} \equiv 1$ "

$2^1, 2^2, \ldots, 2^{10} \equiv 2, 4, 8, 5, 10, 9, 7, 3, 6, 1 \pmod{11}$

and after that it repeats because

$2^{10} \equiv 1 \pmod{11} \longrightarrow 2^{10+k} \equiv 2^k \pmod{11}$

$2^{100} = 2^{10 \cdot 10} = (2^{10})^{10} \equiv 1^{10} = 1 \pmod{11}$

$\boxed{z = 1}$

11.14. *Find all integers* $x \in [0, 12)$ *satisfying the modular equation* $5x \equiv 10 \pmod{12}$*.* *[Hint 183]*

$x \equiv 2 \pmod{12}$ {since $\gcd(5, 12) = 1$}

so in $x \in [0, 12)$, $\boxed{x = 2}$

11.15. *Find all integers* $x \in [0, 12)$ *satisfying the modular equation* $5x \equiv 11 \pmod{12}$. *[Hint 140]*

$5(5x) \equiv 5.11 \pmod{12}$

$25x \equiv x \pmod{12}$

$x \equiv 7 \pmod{12}$

So in the $[0,12]$, then $\boxed{x = 7}$

11.16. *Find all integers* $x \in [0, 12)$ *satisfying the following modular equations (one at a time, not all at once):*

a. $2x \equiv 10 \pmod{12}$ $\quad x \equiv 5 \pmod 6$ – $x = 5, 11$

b. $2x \equiv 8 \pmod{12}$ $\quad x \equiv 4 \pmod 6$ – $x = 4, 10$

c. $2x \equiv 9 \pmod{12}$ $\quad 2x - 9 = 12k$, $\underbrace{2x}_{\text{even}} = \underbrace{\underbrace{9}_{\text{odd}} + \underbrace{12k}_{\text{even}}}_{\text{odd}}$ no Solution

d. $10x \equiv 10 \pmod{12}$ $\quad x \equiv 1 \pmod 6$ – $x = 1, 7$

e. $10x \equiv 8 \pmod{12}$ $\quad 5x \equiv 4 \pmod 6 \leftrightarrow -x \equiv 4 \pmod 6 \leftrightarrow x \equiv 2 \pmod 6$

$x = 2, 8$

f. $12x \equiv 12 \pmod{12}$ $\quad x \equiv 1 \pmod 1$ – $x = 0, 1, 2, 3, \ldots, 12$

g. $12x \equiv 11 \pmod{12}$ $\quad 12x - 11 = 12k$, $\underbrace{12x}_{\text{even}} = \underbrace{\underbrace{12k}_{\text{even}} + \underbrace{11}_{\text{odd}}}_{\text{odd}}$ no Solution

11.17. *Use the method given in the proof of the Chinese Remainder Theorem (Theorem 11.8) to solve the linear modular system* $\{x \equiv 5 \pmod 9, x \equiv 1 \pmod{11}\}$. *[Hint 51]*

* For integers a, n, an inverse of a modulo n is an integer a^{-1} such that $a.a^{-1} \equiv 1 \pmod n$

$\begin{cases} x \equiv 5 \pmod 9 \\ x \equiv 1 \pmod{11} \end{cases}$ So, $m = 9$, $a = 5$, $n = 11$, $b = 1$

we need m' with $9m' \equiv 1 \pmod{11}$

$\overset{m'}{9}.\overset{b}{5} = 45 \equiv 1 \pmod{11}$

we need n' with $11n' \equiv 1 \pmod 9$

$2.\underset{n'}{5} = 10 \equiv 1 \pmod 9$

$x_0 = bmm' + ann'$

$x_0 = 1(9)(5) + 5(11)(5) = 45 + 275 = 320$

$320 \bmod 99 = 320 - 3.99 = 23$, $x = 23$. $(x = 23 + 99k, k \in \mathbb{Z})$

$x \equiv 23 \pmod{99}$

11.18. *Use the method given in the proof of the Chinese Remainder Theorem (Theorem 11.8) to solve the linear modular system* $\{x \equiv 5 \pmod 9, x \equiv -5 \pmod{11}\}$.

$$\begin{cases} x \equiv 5 \pmod 9 \\ x \equiv -5 \pmod{11} \end{cases}$$

$$\begin{cases} x \equiv 5 \pmod 9 \\ x \equiv 6 \pmod{11} \end{cases}$$

$m = 9 \qquad a = 5$

$n = 11 \qquad b = 6$

$m' = 5 \qquad n' = 5$

$x_0 = bmm' + ann'$

$x_0 = 545$

$mn = 99$

$545 \bmod 99 : \ 545 - 4 \cdot 99 = 50$

$x \equiv 50 \pmod{99}$

11.19. *Let* $m, n \in \mathbb{N}$, *and set* $d = \gcd(m, n)$. *Let* $b \in \mathbb{Z}$ *with* $d \nmid b$. *Prove that* $mx \equiv b \pmod n$ *has no solutions.* *[Hint 160]*

$n \mid (mx - b)$

$mx - b = kn \qquad b = mx - kn$

$\gcd(m,n) = 1 \quad d \mid (mx - kn) \longrightarrow d \mid b$ Contradiction

$d \mid m \rightarrow d \mid mx$

$d \mid n \rightarrow d \mid kn$

There for our assumption that a solution x exists is false.

So $mx \equiv b \pmod n$ has no solution.

11.20. *Let* $m, n \in \mathbb{N}$, *and set* $d = \gcd(m, n)$. *Let* $b \in \mathbb{Z}$ *with* $d \mid b$. *Prove that* $mx \equiv b \pmod n$ *has exactly* d *solutions in* $[0, n)$. *[Hint 277]*

$d = \gcd(m,n)$, $m = dm_0$, $n = dn_0$ with $\gcd(m_0, n_0) = 1$

Since $d \mid b$, $b = db_0$, for some integer b_0

$mx \equiv b \pmod n \longleftrightarrow dm_0 x \equiv db_0 \pmod{dn_0}$

$m\, m_0^{-1} \equiv 1 \pmod{n_0}$

$x \equiv m_0^{-1} b$

ch. 11.3

Partitions: Partition of set of S is a collection of nonempty sets $\{S_1, S_2, \ldots, S_k\}$ such that: ① cover: $S_1 \cup S_2 \cup \ldots \cup S_k = S$

② No overlap: if $i \neq j$, then $S_i \cap S_j = \emptyset$

Equivalence class:

we start by we already have an equivalence relation R on S that means R is:

- Reflexive: xRx for all x
- symmetric: $xRy \to yRx$
- transitive: xRy and $yRz \to xRz$

For each element $x \in S$, we, the equivalence class of x is

$[x] = \{y \in S : xRy\}$.

Sometimes written as $[x]_R$ to emphasize which relation

"Take x, grab everything in S that is equivalent to x. That whole blob is $[x]$.

because R is reflexive, we always have xRx, so $x \in [x]$. That is why if you take the union of all classes, you cover the whole set: $\bigcup_{x \in S} [x] = S$

every element sits in at least one equivalence class (its own)

why equivalence classes behave like partition?

Any two equivalence classes are either identical or disjoint

theorem: let R be an equivalence relation on S

take two elements $x, y \in S$, suppose their classes overlap: $[x] \cap [y] \neq \emptyset$

then actually $[x] = [y]$

idea of proof:

Assume there is some element z in both $[x]$ and $[y]$

- $z \in [x]$ means xRz
- $z \in [y]$ means yRz

using symmetry we also have zRx and zRy

now take any $t \in [x]$. then xRt. Combine:

- tRx
- xRz
- zRy

using transivity, every element related to x is also related to y. that gives $[x] \subseteq [y]$. by doing the same arguement and flipping roles of x and y we get $[y] \subseteq [x]$

so $[x] = [y]$

what this means for Partitions? S/R "read S modulo R" or "the quotient of S by R".

$$S/R = \{[x] : x \in S\}$$

Example: even and odd integers (mod 2):

take $S = \mathbb{Z}$, the set of all integers.

let R be the relation "congruent mod 2":

$$a \equiv b \pmod 2 \text{ iff } 2 \mid (a-b)$$

there are exactly two equivalence classes:

1. Evens: all integers with remainder 0 mod 2:

$$\ldots, -4, -2, 0, 2, 4, 6, \ldots$$

2. odds: all integers with remainder 1 mod 2:

$$\ldots, -3, -1, 1, 3, 5, \ldots$$

So the Partition $\mathbb{Z}/\equiv$ for mod 2 is basically:

$$\{\text{evens}, \text{odds}\}$$

Modular Equivalence:

Let $n \in \mathbb{N}$

Define $a \equiv b \pmod n$ as usual $n \mid (a-b)$

The Partion of $\mathbb{Z}$ induced by the relation is

$$(\mathbb{Z} / \equiv) = \{[0], [1], \cdots, [n-1]\}$$

"there are exactly n equivalence classes, one for each remainder $0, 1, \ldots, n-1$"

why?

- by devision alghorithm, any integers x can be uniquely written as:

$$x = qn + r \quad , 0 \leq r < n$$

- two integers a, b are congruent mod n iff they have the same remainder.

So

- the class $[0]$ = all integers with remainder 0 mod n

⋮

- the class $[n-1]$ = all integers with remainder $n-1$ mod n.

Any integers lies exactly in one of those classes and there are no others.

Ex) $n = 3$

- $[0] = \{\ldots, -6, -3, 0, 3, 6, \ldots\}$
- $[1] = \{\ldots, -5, -2, 1, 4, 7, \ldots\}$
- $[2] = \{\ldots, -4, -1, 2, 5, 8, \ldots\}$

So $\mathbb{Z} / \equiv \pmod 3 = \{[0], [1], [2]\}$

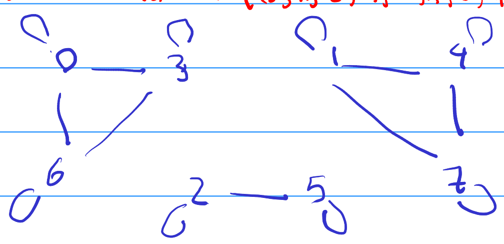
graph:

Example: graph for "$\equiv$ mod 3" restricted to $S = \{0, 1, 2, 3, 4, 5, 6, 7\}$

class $[0]$: $\{0, 3, 6\}$

class $[1]$: $\{1, 4, 7\}$

class $[2]$: $\{2, 5\}$



11.21. *Let* $S = \{0, 1, 2, \ldots, 19\}$. *Draw the graph for equivalence, modulo 3, on* S. *[Hint 123]*

$C_0 = \{0, 3, 6, 9, 12, 15, 18\}$

$C_1 = \{1, 4, 7, 10, 13, 16, 19\}$

$C_2 = \{2, 5, 8, 11, 14, 17\}$

11.23. *Consider the equivalence relation from exercise 11.1. Find* $[0.4]$; *give this in set-builder notation, without any direct reference to* R. *[Hint 164]*

relation from 11.1:

on $\mathbb{Q}$:

$(a, b) \in R \longleftrightarrow a - b \in \mathbb{Z}$

$[0.4]$

$[0.4] = \{q \in \mathbb{Q} : q \sim 0.4\}$

$q \sim 0.4 \longleftrightarrow q - 0.4 \in \mathbb{Z}$.

$[0.4] = \{q \in \mathbb{Q} : q - 0.4 \in \mathbb{Z}\} = \{0.4 + n : n \in \mathbb{Z}\}$

11.24. *Consider the equivalence relation from exercise 11.3. Find* $[x^2 + 3x + 1]$; *give this in description notation, without any direct reference to* R. *[Hint 114]*

Relation on exercise 11.3:

$(p(x), q(x)) \in R$ iff $p(0) - q(0) = 0$ i.e. $p(0) = q(0)$

$[x^2 + 3x + 1] = \{p(x) \in \mathbb{Z}[x] : p(0) = 1\}$

11.25. *Consider the equivalence relation from exercise 11.5. Find* $[x^2 + 3x + 1]$; *give this in description notation, without any direct reference to* R. *[Hint 119]*

$(p(x), q(x)) \in R$ iff $p'(x) = q'(x)$

$[x^2 + 3x + 1] = \{x^2 + 3x + c : c \in \mathbb{Z}\}$ $p'(x) = 2x + 3$

11.26. *Let* $n \in \mathbb{N}$, *and let* $\equiv$ *be the equivalence relation modulo* n, *on* $\mathbb{Z}$. *Let* $a, b \in \mathbb{Z}$. *Define* $[a] + [b] = \{x + y : x \in [a], y \in [b]\}$. *Prove that* $[a] + [b] = [a + b]$. *(equal as sets)* [Hint 73]

equality of sets mean : $[a]+[b] \subseteq [a+b]$
$[a+b] \subseteq [a]+[b]$

$[a]+[b] \subseteq [a+b]$:

take any element $z \in [a]+[b]$
by definition of that set there exist some integer $x \in [a]$ and $y \in [b]$, such that $z = x+y$
because if $x \in [a]$, $x \equiv a \pmod{n}$
if $y \in [b]$, $y \equiv b \pmod{n}$
$x \equiv a \pmod{n}$ $y \equiv b \pmod{n}$
$x+y \equiv a+b \pmod{n}$ but also $z = x+y$ so
$z \equiv a+b \pmod{n}$, $z \in [a+b]$

$[a+b] \subseteq [a]+[b]$: take any $z \in [a+b]$.
$z \equiv (a+b) \pmod{n}$
Let $x = a$, $y = z-a$
$x+y = a+(z-a) = z$
• $x = a$, is clearly in $[a]$, because $a \equiv a \pmod{n}$
For y:
• we know $z \equiv a+b \pmod{n}$
$z - a \equiv b \pmod{n}$
$y \equiv b \pmod{n}$
$y \in [b]$

11.27. *Let* $n \in \mathbb{N}$*, and let* $\equiv$ *be the equivalence relation modulo* n*, on* $\mathbb{Z}$*. Let* $a, b \in \mathbb{Z}$*. Define* $[a] \cdot [b] = \{t \in \mathbb{Z} : \exists x \in [a], \exists y \in [b], t \equiv xy\}$*. Prove that* $[a] \cdot [b] = [a \cdot b]$*. (equal as sets)* [Hint 52]

$[a] \cdot [b] \subseteq [ab]$

take any $t \in [a] \cdot [b]$

by definition there exist some $x \in [a]$ and $y \in [b]$ such that

$t \equiv xy \pmod n$

$x \in [a]$, $x \equiv a \pmod n$

$y \in [b]$, $y \equiv b \pmod n$

multiplication rule for congruences: $xy \equiv ab \pmod n$

$t \equiv ab \pmod n$

$t \in [ab]$, because $t \equiv ab \pmod n$

$[ab] \subseteq [a] \cdot [b]$

by definition of $[ab]$, $t \equiv ab \pmod n$

- Let $x = a$, then I already have $x \in [a]$ since $x \equiv x \pmod n$
- Let $y = b$, $y \in [b]$, since $y \equiv y \pmod n$

$xy = ab$

we already know: $t \equiv ab \pmod n$

$t \equiv ab = xy \pmod n$

So, there exists $x \in [a], y \in [b]$ (namely $x = a, y = b$) with $t \equiv xy \pmod n$

ch 12.1:

Partial order / Poset: we have a set S and a relation R on S, R is partial order if it has these three properties:

Reflexive: for every $x \in S$, we have xRx (everything is related to itself)

Antisymmetric: for every $x, y \in S$, if we have xRy and Ryx, then $x=y$

(You cannot have two different things each "less than equal to" each other)

Transitive: for any $x, y, z \in S$, if xRy and yRz, then xRz.

(order doesn't break when you go to middle element)

when we put a set together with a partial order, we call it partially order set or Poset, we usually write as (S, R).

Ex) $S = \mathbb{N}$ and $R = \leq$

so $(\mathbb{N}, \leq)$ is poset

comparable vs incomparable: take a poset (S, R) and two distinct elements $a, b \in S$.

- we call a, b comparable if either aRb or bRa

  (one is $\leq$ other in that order)

- if neither aRb nor bRa holds, they are incomparable or parallel, written $a \| b$.

if every pair of distinct element is comparable, then the relation is total order (or linear order)

- Ex: the usual $\leq$ on $\mathbb{Z}$ is total: any two integers one is $\leq$ the other.

total order: everyone can be lined up

partial order; some elements are unrelated in the order.

Ex) (a) Divisibility on $\mathbb{N}$

- set $S = \mathbb{N}$
- Relation: aRb iff $a|b$

incomparability example) 3 and 4:

$3 \nmid 4$ and $4 \nmid 3$, so $3 \| 4$

Reflexive: each number divides itself

Antisymmetric: if $a|b$ and $b|a$, then $a=b$

transitive: if $a|b$ and $b|c$, then $a|c$

so divisity is a partial order

Ex) (b) Subset relation on power set:

- Take any set T. Let $S = 2^T$ (all subsets of T)
- ARB iff $A \subseteq B$ — reflexive, antisymmetric, transitive.

Ex) $T = \{1, 2, 3\}$. then 2^T has 8 subsets.

$\{1\} \subseteq \{1, 2\}$, but $\{1\}$ and $\{2\}$ are incomparable.

Hasse diagram - How to draw posets:

Basic rules: Vertices: 1. elements of the set S

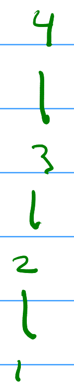
2. you draw an edge between x, y only if: $x < y$ in the partial order, and there is no element z with $x < z < y$

3. the diagram is drawn with vertical orientation: if there is an edge from x to y, then y is drawn higher on the page than x

4. $x \le x$ is omitted because we already know the relation is reflexive

5. Edges implied by transitivity are also omitted.

Ex) (1)

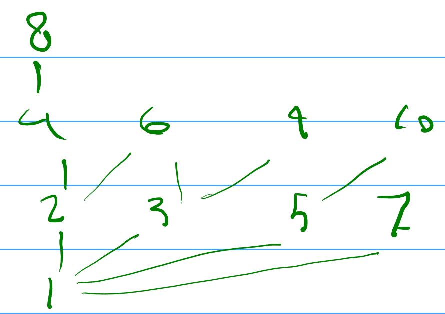


$\le$ on $\{1, 2, 3, 4\}$

Ex) (2) 1 2 3 4

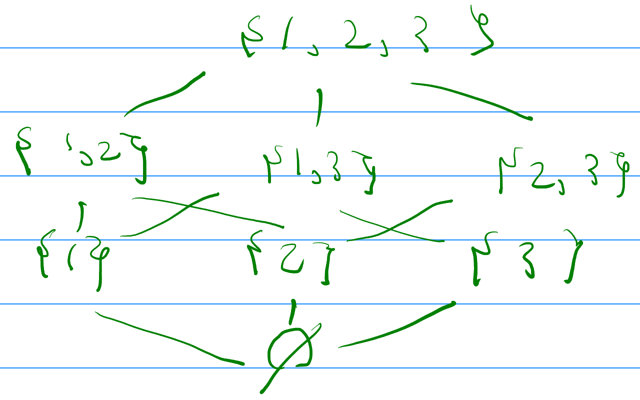
R diagonal on $\{1, 2, 3, 4\}$

Ex) (3)



$|$ on $\{1, 2, 3, \ldots, 10\}$

Ex) (4):



$\subseteq$ on $2^{\{1,2,3\}}$

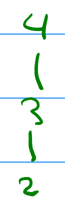
Interval Poset: Let R be a Partial order on S, and Pick $a, b \in S$, with aRb (so $a \leq b$ in this order)

The interval $[a,b]$ with respect to R is:

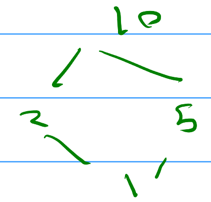
$$[a,b] = \{ x \in S : aRx \text{ and } xRb \}$$

So all elements between a and b. then we take that subset and keep the same order R, but restricated to those elements.

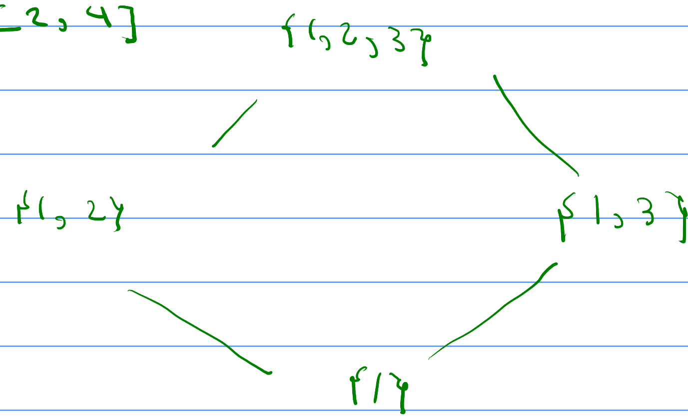
Ex) (a):



$\leq$ on $\mathbb{N}$ restricited to $[2,4]$



$|$ on $\mathbb{N}$ restricated to $[1,10]$



$\subseteq$ on 2^n, restricated to $[\{1\}, \{1,2,3\}]$

12.1. *Let* $S = \mathbb{Z}$*, and let* R *be the relation of divisibility,* $|$*. Prove that* R *is not a partial order.* [Hint 65]

R is a Partial order iff it is → reflexive ✓
↳ Anti symmetric
↳ transitive ✓

we can use counter example for Anti symmetry

let $1 | -1$ $\quad -1 = 1 \cdot (-1)$
$-1 | 1$ $\quad 1 = (-1)(-1)$

we have $1 | -1$ and $-1 | 1$ but $1 \neq -1$

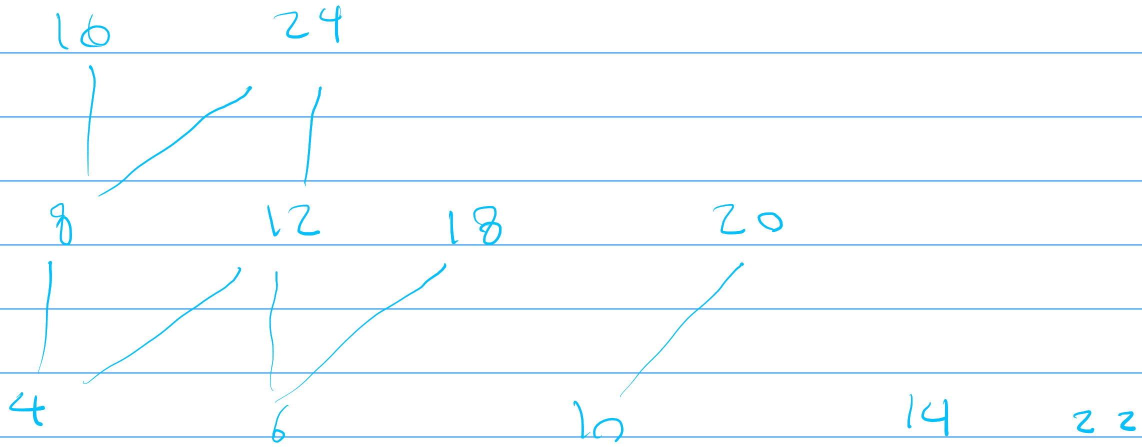
so the divisibility relation on all integers $\mathbb{Z}$ is not Partial order.

12.2. *Prove that the three relations in Example 12.3 are partial orders.* [Hint 59]

12.3. *Draw the Hasse diagram for the relation* $|$ *on* $S = \{4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24\}$*.* [Hint 185]

(a) Divisibility $|$ on $\mathbb{N}$: let $a, b, c \in \mathbb{N}$, Reflexive: $a | a$ because $a = a \cdot 1$
Antisymmetric: if $a | b$, $b | a$, then there are $m, n \in \mathbb{N}$ with $b = am$, $a = bn$
$a = (am)n = a(mn)$, so $a(mn - 1) = 0$, so in $\mathbb{N}$ $mn = 1$, since $m, n \in \mathbb{N}$, that forces $m = 1, n = 1$ and therefore $a = b$
Transitive: if $a | b$ and $b | c$, $b = ak$, $c = bl$, $c = bl = (ak)l = a(kl)$
So $a | c$.

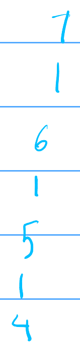
(b) Subset: $A, B, C \subseteq T$, Reflexive: $A \subseteq A$ is always True, Antisymmetric: if $A \subseteq B$ and $B \subseteq A$, then every element of A is in B and every element of B is in A so $A = B$, Transitive: if $A \subseteq B$ and $B \subseteq C$, Then every element of A is in B and every element of B is in C, so $A \subseteq C$. So $(2^T, \subseteq)$ is a Partial order.



12.4. *Draw the Hasse diagram for the diagonal relation on* $S = \{x, y, z\}$. [Hint 266]

$x \quad y \quad z$

12.5. *Consider the relation* $\leq$ *on* $\mathbb{N}$. *Draw the Hasse diagram for the interval poset* $[4, 7]$. [Hint 16]



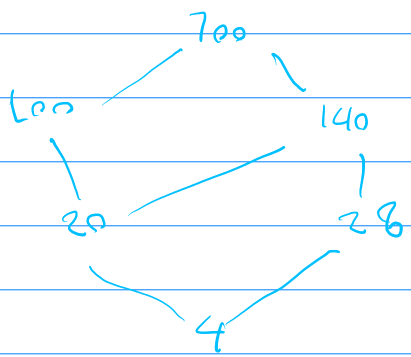
12.6. *Consider the relation* $\mid$ *on* $\mathbb{N}$. *Draw the Hasse diagram for the interval poset* $[4, 700]$. [Hint 101]

$700 = 2^2 \cdot 5^2 \cdot 7$

$\{x \in \mathbb{N} : 4 \mid x \text{ and } x \mid 700\}$

$4 = 2^2$

$x = 2^2 \cdot 5^a \cdot 7^b$



$a \in \{0, 1, 2\}$ and $b \in \{0, 1\}$

$a = 0, b = 0 \to 4$, $a = 1, b = 0 \to 20$, $a = 2, b = 0 \to 100$

$a = 0, b = 1 \to 28$, $a = 1, b = 1 \to 140$, $a = 2, b = 1 \to 700$

12.2

Greatest, maximal and upper bound: Poset (S,R), some subset $T \subseteq S$, big elements of T:

a) Greatest element (a.k.a maximum):

"a is greatest in T") means:

1. $a \in T$, and
2. for every $x \in T$, we have $x R a$

So every element of T is below a in the poset.

- In $(\{1,2,\dots,10\}, \leq)$, the set $T = \{1,2,3,4,5\}$ has greatest element 5
- In divisibility poset on $S = \{1,\dots,10\}$ with $|$, the subset $T = S$ has no greatest element: no single number is divisible by all the others.

If a greatest element exist it is unique (because if two element are each $\geq$ the other, antisymmetry forces them to be equal.

b) Maximal element:

"a is maximal in T" means:

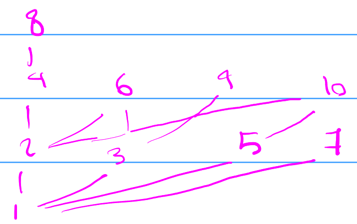
1. $a \in T$, and
2. there is no $x \in T$ with $a \neq x$ and aRx "nothing in T is strictly above it"

Differences from greatest:

- There can be many maximal elements
- A greatest number if it exists is always maximal, but not vice versa.

Example) divisibility on $\{1,\dots,10\}$:

Poset:



Maximal: 8 divides no larger number in $\{1,\dots,10\}$ so 8 is maximal

But there is no greatest element

c) upper bound:

"a is upper bound of T" means:

- For all $x \in T$, xRa - but a doesn't have belong T.

so upper bound lives in the whole poset S, not necessary subset

Example)

- Take $(\mathbb{R}, \leq)$ and $T = (0, 3)$ = all numbers strictly between 0 and 3.
- there is no greatest element in T (every number in T has something bigger still inside T.
- but there are lots of upper bounds: $3, 5, 10, 10^6, \cdots$ any number ≥ 3 is an upperbound of T.

So:

- Greatest in T: upper bound that also lies in T, and is minimal among upper bounds
- upper bound = just "on or above everything in T", maybe outside T

we have analogous definitions for 'least', 'minimal', and 'lower bound' by flipping the order (down instead of up)

least, minimal, and lower bounds: same story but downwards

a) least element (a.k.a minimum)

"a is least in T" if

1. $a \in T$, and
2. for all $x \in T$, aRx

b) Minimal element:

"a is minimal in T" if

1. $a \in T$ and
2. there is no $x \in T$ with $x \neq a$ and xRa

c) Lower bounds:

"a is lower bound of T" if

- For all $x \in T$, aRx, but a might be outside of T.

well ordered sets (recall):

a partial order R on S is well ordered if:

- the order is total, and
- Every non empty subset $T \subseteq S$ has a least element.

Examples:

- $(\mathbb{N}, \leq)$ is well ordered.
- $(\mathbb{Z}, \leq)$ is not (subset $\mathbb{Z}$ has no least element)
- $(\mathbb{R}, \leq)$ is not ($(0,3)$ as an counter example).

Note: every well ordered is total order, but not every total order is well ordered.

Product and lexicographic orders:

Now we have two posets: (S_1, R_1) and (S_2, R_2), we want a partial order on pairs (a,b) in the Cartesian product $S_1 \times S_2$

a) lexicographic (lex) order:

we say $(a,b) R (c,d)$ in lex order if:

1. $a \neq c$ and $a R_1 c$, or
2. $a = c$ and $b R_2 d$.

So first compare first coordinates:

- if they differ, the one with smaller first cordinate is smaller overall
- if first coordinates are equal, you break ties by comparing second cordinate)

if both R_1 and R_2 are total orders, lex order is also a total order.

b) Product order:

we say $(a,b) \leq (c,d)$ in the product order if:

- $a R_1 c$ and $b R_2 d$

Example) Take $S_1 = S_2 = \mathbb{N}$ with usual $\leq$.

- in product order $(1,5) \leq (3,7)$, because $1 \leq 3$, and $5 \leq 7$
- but $(1,5)$ and $(3,3)$ are incomparable.

Partial orders on $\mathbb{C}$ (complex numbers)

Lex order: compare by real part first, break ties with imaginary part

Product order: $(a+bi) \le (c+di)$ if $a \le c$ and $b \le d$

even though we can cook up partial order (or total orders) on $\mathbb{C}$ there is no total order on $\mathbb{C}$ that respect the algebra in the same nice way that as $\le$ does on $\mathbb{R}$.

Extending partial orders & linear extensions:

Now we have two partial order on the set S, call them R_1 and R_2

we say "R_2 extends R_1" if:

- As sets of ordered pairs, $R_1 \subseteq R_2$.

that is every relation that holds under R_1, also holds under R_2

Equivalently:

For all $a, b \in S$, if aR_1b then aR_2b.

so R_2 never contradicts R_1; it just adds more comparabilities.

Linear extention:

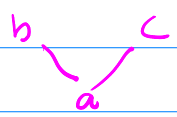
if R_2 extends R_1 and R_2 is total order on S, then R_2 is called a linear extention of R_1

"Linear" = you can write the elements of S in a line:

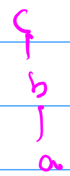
$$x_1 < x_2 < \cdots < x_n$$

And it respects the original partial order (any $a \le_1 b$ appears with a before b in the line)

Ex)



has two extentions



and

Order Extention Principle & Scheduling:

Order Extention Principle: let R be a partial order on an (infinite) set S, then R has at least one linear extention.

Scheduling and Topological sort:

imagine S is set of tasks, and aRb means "task a must be done before task b"

- These precedence constraints from a partial order.
- A schedule that respects all prerequisites is exactly a linear extention of this partial order.

So topological sort = linear extention.

12.7. *Consider the relations from Exercises 12.3, 12.4, 12.5, 12.6. Identify any greatest, maximal, least, minimal elements.* [Hint 96]

12.3: no least element

no greatest element

minimal: $\{4, 6, 10, 14, 22\}$

maximal: $\{14, 16, 18, 20, 22, 24\}$

12.4: no least element

no greatest element

minimal $= \{x, y, z\}$

maximal: $\{x, y, z\}$

12.5: Greatest: 7

Least: 4

minimal: 4

maximal: 7

12.6: least: 4

Greatest: 700

minimal: 4

maximal: 700

12.8. *Let* R *be a partial order on set* S, *and* $T \subseteq S$. *Suppose that* a *is greatest in* T. *Prove that* a *is maximal in* T. [Hint 47]

a is greatest in T means: 1. $a \in T$, and 2. for every $x \in T$, we have xRa

a is maximal in T means: 1. $a \in T$, and 2. There is no $x \in T$ with $x \neq a$ and aRx

Proof:

Assume R is Partial order on S, $T \subseteq S$, and a is greatest in T so we know $x \in T$, and for every $x \in T$, xRa. To show maximal we need to show there is no $x \in T$ with $x \neq a$ and aRx. So argue by contradiction:

Suppose there is some $x \in T$ such that aRx and $x \neq a$ but since a is greatest in T, condition above says for every $x \in T$ we have xRa, in particular for this same x we also have xRa so we have both xRa and aRx, because R is Partial order it is antisymmetric which states if aRx and xRa then $a = x$ so from aRx and xRa we must conclude $a = x$, but that contradicts our assumption $x \neq a$ therefor our assumption is impossible so there is no $x \in T$ with $x \neq a$ and aRx

12.9. *Let* R *be a partial order on set* S, *and* $T \subseteq S$. *Suppose that* $a, a' \in T$ *are both greatest in* T. *Prove that* $a = a'$. [Hint 293]

because a' is greatest in T and $a \in T$ $a R a'$ (since all elements of T are $\leq a'$) and since a is greatest and $a' \in T$ $a' R a$, so we have $a R a'$ and $a' R a$ by using antisymmetry of partial order we know $a = a'$

12.12. *Consider the partial order* | *on* $\mathbb{N}$, *and set* $T = \{6, 10\}$. *Find three elements* $a, a', a'' \in \mathbb{N}$ *such that* $a|a'$, $a \parallel a''$, $a' \parallel a''$ *and each of* a, a', a'' *are upper bounds for* T. [Hint 288]

to be upper bound for T it might satisfy $6|u$ and $10|u$. so u is a common multiple of 6 and 10. This means u is a multiple of $\text{lcm}(6,10) = 30$

so every upper bound of T is of the form $30k$ for some $k \in \mathbb{N}$

lets take $a = 60$ $a' = 120$ $a'' = 90$

$6|60$ $10|60 \to 60$ is an upper bound

$6|120$ $10|120 \to 120$ is an upper bound

$6|90$ $10|90 \to 90$ is an upper bound

we then check comparabilities.

1. $a|a' \to 60|120$

2. $a \parallel a'$ (incomparable)

$60 \nmid 90$
$90 \nmid 60$ } neither devides the other therefor incomparabable

3. $a' \parallel a''$

$120 \nmid 90$
$90 \nmid 120$ } again neither devides the other.

So $a = 60$, $a' = 120$, $a'' = 90$ it satisfies.

12.13. *Consider the partial order* $\mid$ *on* $\mathbb{N}$, *and set* $T = \{6, 10\}$. *Find three elements* $a, a', a'' \in \mathbb{N}$ *such that* $a \parallel a'$, $a \parallel a''$, $a' \parallel a''$ *and each of* a, a', a'' *are upper bounds for* T. [Hint 230]

we want all $a \nmid a'$ $a \nmid a''$ and $a' \parallel a''$ if $a = 30k_1$ and $a' = 30k_2$ and $a'' = 30k_3$

we need to chose k_1, k_2 and k_3 such that none of them devides another

Simple choice: $k_1 = 2, k_2 = 3, k_3 = 5$ so I have $a = 60, a' = 90, a'' = 150$

12.14. *Let* R *be a partial order on set* S, *and let* $a, b \in S$ *with* aRb. *Prove that the interval poset* $[a, b]$ *has a greatest and a least element.* [Hint 248]

$[a,b] := \{x \in S : aRx \text{ and } xRb\}$

1. showing a is the least element of $[a,b]$:

(i) $a \in [a,b]$: because R is a Partial order, it is reflexive aRa and we are given aRb, so both conditions are met and $a \in [a,b]$.

(ii) for any $x \in [a,b]$ we have, aRx:

if $x \in [a,b]$ we know aRx and xRb, in Particular we already have aRx.

2. showing b is greatest element of $[a,b]$:

$T = [a,b]$

(i) $b \in [a,b]$:

we are given aRb

by Reflexivity of R, bRb

so $b \in [a,b]$

(ii) if $x \in [a,b]$ we know aRx and xRb

in Particular xRb holds.

12.15. *Let* R *be a relation on* S, *and let* a, b ∈ S *with* aRb. *Suppose that* R *is a total order. Prove that the interval poset* [a, b] *is a total order.* [Hint 210]

$R_{[a,b]} := R \cap ([a,b] \times [a,b])$. So $(x,y) \in R_{[a,b]}$ mean $x,y \in [a,b]$ and xRy

Reflexive: Take any $x \in [a,b]$, because R is reflexive on S, we have xRx since both entries are $x \in [a,b]$, $(x,x) \in R_{[a,b]}$

Antisymmetric: suppose $x, y \in [a,b]$ and $x R_{[a,b]} y$ and $y R_{[a,b]} x$. because R is antisymmetric on all of S, we must have $x=y$

Transitive: Suppose $x, y, z \in [a,b]$ and $x R_{[a,b]} y$ and $y R_{[a,b]} z$. Since R is transitive on S, xRz with $x, z \in [a,b]$ we get $(x,z) \in R_{[a,b]}$

So is Poset.

the Partial order is total if any two elements in the ground set are comparable. (For all x,y in the set, either xRy or yRx. Now take $x,y \in [a,b]$

Since $[a,b] \subseteq S$ and R is total order on S:

either xRy or yRx

Since $x,y \in [a,b]$ whichever of those pairs holds is actually in the restricted relation $R_{[a,b]}$.

12.16. *Let* R *be a well-order on* S. *Prove that* R *is a total order.* [Hint 220]

Take any elements $a,b \in S$, consider the subset $T = \{a, b\} \subseteq S$, it is non empty so because R is well ordered, T has a least element call that least element $c \in T$, so c is either a,b and least in T mean $\forall x \in T$, cRx

because $c \in \{a,b\}$, there are two Possibilities:

Case 1: $c=a$ so we have aRa and aRb (since $b \in T$), a and b are comparable

Case 2: $c=b$ so we have bRa and bRb, so and b are comparable

In every case for any a,b in S, we get either aRb or bRa

12.18. *Let* R_1, R_2 *both be the usual order* $\leq$ *on* $\mathbb{N}$. *Let* $T = \{(1,3), (2,2), (4,1)\}$. *Identify any greatest and maximal elements in* T, *in the lex order on* $\mathbb{N} \times \mathbb{N}$. *[Hint 156]*

Compare all Pairs in lex order:

$$\left.\begin{array}{l} (1,3) <_{lex} (2,2) \\ (2,2) <_{lex} (4,1) \\ (1,3) <_{lex} (4,1) \end{array}\right\} (1,3) <_{lex} (2,2) <_{lex} (4,1)$$

greatest element : $(4,1)$. maximal element : $(4,1)$.

12.19. *Let* R_1, R_2 *both be the usual order* $\leq$ *on* $\mathbb{N}$. *Let* $T = \{(1,3), (2,2), (4,1)\}$. *Identify any greatest and maximal elements in* T, *in the product order on* $\mathbb{N} \times \mathbb{N}$. *[Hint 265]*

$(1,3)$, $(2,2)$ incomparable greatest element : none

$(1,3)$, $(4,1)$ incomparable Maximal element: $(1,3)$, $(2,2)$, $(4,1)$

$(2,2)$, $(4,1)$ incomparable

12.20. *Find three partial orders on* $\mathbb{C}$; *one in which* $1 - i < 1 + i$, *one in which* $1 - i > 1 + i$, *and one in which* $1 - i \parallel 1 + i$. *[Hint 21]*

$(1,-1) <_{lex} (1,1)$

or $1-i < 1+i$

let have backword lexicographic Partial order as ver then $(1,-1) > (1,1)$

or $1-i > 1+i$

let have order by length : $|a+bi| = \sqrt{a^2+b^2}$ so $|1-i| = 2$

$|1+i| = 2$

but $1-i \not< 1+i$

so $1-i \parallel 1+i$

12.3.

Chain and antichain: we start with a poset: a set S with a partial order R on it.

Chain: A subset $T \subseteq S$ is a chain if every two elements of T are comparable

- "Comparable": means for any $x, y \in T$, either xRy or yRx (or both if they are equal)
- In other words, the relation R is restricated to T is a total order.

Example)

- In $(\mathbb{N}, \leq)$, any subset like $\{2, 5, 10, 100\}$ is a chain because for any two numbers you can say which is $\leq$ which
- In divisibility poset on $\{1, \dots, 10\}$ (edges $a|b$), $\{1, 2, 4, 8\}$ is a chain since $1|2|4|8$

Anti chain:

A subset of $T \subseteq S$ is an antichain if no distinct elements are comparable.

- for distinct $x, y \in T$, neither xRy nor yRx is true.
- R is restricated to T is the diagonal relation only.

So antichain is set of points in hasse diagram where none are above or below any other.

Example) divisibility on $\{1, \dots, 10\}$:

$\{6, 7, 8, 9, 10\}$ is an antichain.

Height and width of a poset: Now we want to measure how "tall" and how "wide" a poset is.

- Height of R: $\text{height}(R) = \max\{|T| : T \subseteq S \text{ is a chain}\}$.
  so size of the longest chain.
- Width of R: $\text{width}(R) = \max\{|T| : T \subseteq S \text{ is an antichain}\}$.
  So size of the largest antichain.

Example) divisibility on $\{1, \dots, 10\}$:

A longest chain is $1, 2, 4, 8$, so max length is 4 so height is 4.

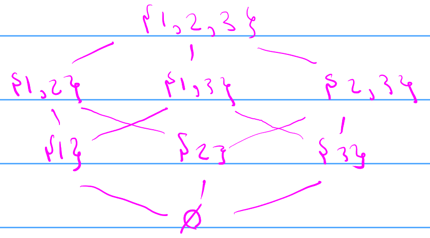
the longest antichain is 5 (i.e $\{6, 7, 8, 9, 10\}$) so width is 5.

Subset Poset(2^T, $\subseteq$) and Sperner's theorem

lets look at special case where all subsets of a final set T ordered by inclusion.

- Ground set: 2^T (all subsets of T)
- order: $\subseteq$

if $|T|=3$ with $T=\{1,2,3\}$, Hasse digram would look like:



level: "level k" is all subsets of T with exactly k elements

for example level 2 in $2^{\{1,2,3\}}$ is $\{\{1,2\},\{1,3\},\{2,3\}\}$

* each level is an antichain

* different levels can be linked by inclusion.

- Hieght of this poset: longest chain from $\emptyset$ to T, has length $n+1$ if $|T|=n$
- width: how big can antichain be?

by taking one entire level we get an anti chain, so level k has $\binom{n}{k}$ subsets but the big question is: which level is largest? The middle level(s), around $k=\lfloor n/2 \rfloor$

Sperner's Theorem:

For the poset $(2^T, \subseteq)$ with $|T|=n$ the width is

$$\binom{n}{\lfloor n/2 \rfloor}$$

i.e. the size of middle level.

Dilworth's Theorem: There is a deep connection between chains and anti chains.

Chain decomposition / Partition: a chain partition of S is a way to split S into disjoint chains: $S = S_1 \cup \dots \cup S_k$ where each S is chain and they don't overlap.

Dilworth's Theorem says:

if you partition S into chains $S_1, \dots, S_k$ then width $\leq k$

Examples) Divisibility on $\{1, \dots, 10\}$ you can partition into 5 chains, e.g:

- $\{1, 2, 4, 8\}$
- $\{3, 6\}$
- $\{9\}$
- $\{5, 7\}$
- $\{10\}$

By Dilworth algorithm:

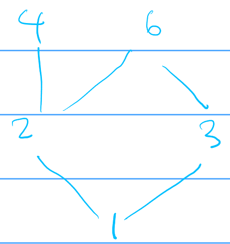
width ≥ 5 (Because that is minimal if partition is optimal)

Zorn's lemma: let R be a partial order on infinite set S. Suppose that every chain $T \subseteq S$ has an upper bound x_T in S (i.e every element of the chain $\leq x_T$), then S contains a maximal elements.

- upper bound of a set T: element $u \in S$ s.t. for every $t \in T$, tRu
- Maximal element: an element m such that there is no strictly bigger element above it: there is no y with mRy and $m \neq y$

12.21. *Consider the relation | on S = {1, 2, 3, 4, 6}. Find all linear extensions of | on S. (this is really from Chapter 12.2, but that section had too many exercises)* [Hint 286]

1. 1, 2, 3, 4, 6
2. 1, 2, 3, 6, 4
3. 1, 2, 4, 3, 6
4. 1, 3, 2, 4, 6
5. 1, 3, 2, 6, 4



12.22. *Consider the relation | on S = {1, 2, 3, 5, 6}. Find all linear extensions of | on S. (this is really from Chapter 12.2, but that section had too many exercises)* [Hint 207]

1, 2, 3, 5, 6

1, 2, 5, 3, 6

1, 3, 2, 5, 6

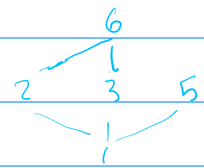
1, 3, 5, 2, 6

1, 5, 2, 3, 6

1, 5, 3, 2, 6

1, 2, 3, 6, 5

1, 3, 2, 6, 5



12.23. *Consider the partial order | on {1, 2, 3, ..., 10}. Without using Dilworth's Theorem, prove that it has no antichain of size 6.* [Hint 212]

$C_1 = \{1, 2, 4, 8\}$

$C_2 = \{3, 6\}$

$C_3 = \{9\}$

$C_4 = \{10\}$

$C_5 = \{5, 7\}$

each C_i is chain and these 5 are disjoint covering all of $\{1, \dots, 10\}$

Let $A \subseteq \{1, \dots, 10\}$ with $|A| = 6$

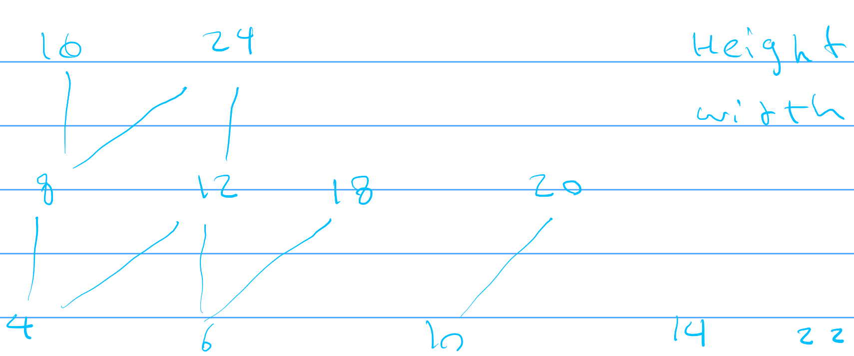
Since the five chains $C_1, \dots C_5$ partition the whole set, every element of A lies exactly in one of these chains. if we put 6 elements into 5 boxes at least one chain must contain two element of A, but then those elements would be comparable so, A is not an antichain.

12.24. *Set* $T = \{a, b, c, d, e\}$. *Find a maximal clutter from* T. *[Hint 72]*

$\binom{5}{2} = 10$

$\{\{a,b\},\{a,c\},\{a,d\},\{a,e\},\{b,c\},\{b,d\},\{b,e\},\{c,d\},\{c,e\},\{d,e\}\}$

12.25. *Consider the relations from Exercises 12.3, 12.4, 12.5, 12.6. Find the height and width of each. Be sure to justify.* *[Hint 285]*



Height = 3

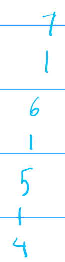
width = 6

12.4. *Draw the Hasse diagram for the diagonal relation on* $S = \{x, y, z\}$. *[Hint 266]*

$x \quad y \quad z$

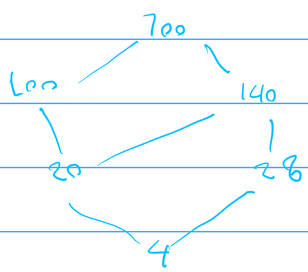
height = 1 width = 3

12.5. *Consider the relation* $\leq$ *on* $\mathbb{N}$. *Draw the Hasse diagram for the interval poset* $[4, 7]$. *[Hint 16]*



height = 4 width = 1

12.6. *Consider the relation* $|$ *on* $\mathbb{N}$. *Draw the Hasse diagram for the interval poset* $[4, 700]$. *[Hint 101]*



height = 4

width = 2

12.26. *Let* R *be a total order on nonempty set* S*. Prove that the width of* R *is* 1*, and the height of* R *is* |S|. [Hint 215]

Because R is total order for any two distinct element $x, y \in S$, we always have either xRy or yRx so every pair of distinct elements are comparable. but in an anti-chain no distinct element are comparable so width$(R) = 1$

Height of a total order is $|S|$, In total order every element of S is comparable, so entire set S itself is a chain. so

There exist a chain of $|S|$, S itself, so height $(R) \geq |S|$

Any chain $T \subseteq S$ is subset of S Therefor cannot be larger than $|S|$

Combining:

height$(R) \geq |S|$ and height$(R) \leq S \rightarrow$ height$(R) = |S|$

12.27. *Let* R_1, R_2 *both be the usual order* $\leq$ *on* $S = \{1, 2, 3\}$*. Let* R *be the product order on* $S \times S$*. Find the width and height of* R*. Be sure to prove your answer.* [Hint 126]

we have $S = \{1, 2, 3\}$ and product order on $S \times S$:

$(a,b) \leq (c,d)$ iff $a \leq c$ and $b \leq d$

So the elements are in 9 pairs:

$(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)$

Height. $(1,1) \leq (1,2) \leq (1,3) \leq (2,3) \leq (3,3)$ hence height ≥ 5

Let $s(i,j) = i+j$, if $(a,b) \leq (c,d)$ then $a \leq c$, $b \leq d$ and at least one equality is strict so $c+d \geq a+b+1$ i.e $s(c,d) \geq s(a,b)+1$

minimum value of s is $s(1,1) = 2$ and maximum vale of s is $s(3,3) = 6$

So at most we have $6-2+1 = 5$ so no chain has more than 5 elements and height $= 5$

width: take $A = \{(1,3), (2,2), (3,1)\}$ so width ≥ 3. rows and columns are chain, In antichain you cannot have two comparable elements. so you can have at most one element from each row and at most one element from each column, we have 3 rows and therefor if you had 4 elements 2 of them would lie into same row, so width ≤ 3 therefor width $= 3$

13.1

Take $R \subseteq S \times T$, for each pair $(x,y) \in R$, think " x is connected to y"

for each $x \in S$ (vertical lines)

- "Every x has at least one y"

  $\longleftrightarrow$ every vertical line hits the graph at least once.

- " Every x has at most one y"

  $\longleftrightarrow$ every vertical line hits the graph at most once.

Together "at least once" + " at most once" = exactly once

Fuction $f: S \longrightarrow T$:

for every $x \in S$, there exist exactly one $y \in T$ with $(x,y) \in R$.

for each $y \in T$ (horizontal lines)

- " Every y is hit by at least one x"

  $\longleftrightarrow$ every horizontal line hits the graph at least once.

  $\longleftrightarrow$ function is onto / surjective

- " Every y is hit by at most one x "

  $\longleftrightarrow$ every horizontal line hits the graph at most once.

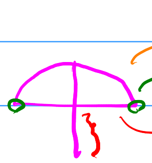
  $\longleftrightarrow$ function is one to one (injective

* A relation $R \subseteq S \times T$ is a function if every x appears exactly once

* A fuction is one to one if no horizontal line hits it twice

* A function is onto if every horizontal line in the codomain range hits at least once.

Here $S = T = [-1,1]$

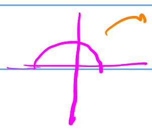


fuction ✓

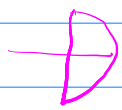
one to one ✗

onto for some y horizontal line doesn't hit at all

$R_1 = \{ (x,y) : x^2 + y^2 = 1, y \geq 0 \}$

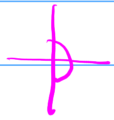


not function from all of S

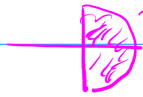


$R_3 = \{ (x,y) : x^2 + y^2 = 1, x \geq 0 \}$

$R_2 = \{ (x,y) : x^2 + y^2 = \frac{1}{4}, y \geq 0 \}$

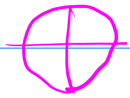


$R_4 = \{(x,y) : x^2+y^2 = \frac{1}{4}, x \geq 0\}$

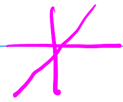


not function from $S \to T$ and $T \to S$

$R_5 = \{(x,y) : x^2+y^2 \leq 1, x \geq 0\}$



$R_6 = \{(x,y) : x^2+y^2 = 1\}$



$R_7 = \{(x,y) : 2+y\}$

13.2: Function: Special type of relation: start with a relation $R \subseteq S \times T$ (pairs (x,y) with $x \in S$, $y \in T$).

R is function from S To T if:

- "Existance: for every $x \in S$ There is at least one $y \in T$ with $(x,y) \in R$"
- "uniqueness: For each $x \in S$ there is at most one such y

So, for for each input x there is exactly one output y.

- $R(x) = y$ or $f(x) = y$
- $R: S \to T$ or $f: S \to T$

Domain: the set you are mapping from, usually denoted S.

Codomain: The set you are mapping into, T in $f: S \to T$

Range/image: The actual output you hit:

$$\text{Im}(f) = \{y \in T : \exists x \in S, f(x) = y\}$$

This is subset of the codomain T.

when two functions are equal?

Two functions are equal if

1. they have the same domain.
2. for every x in that domain, they give the same output

$f(x) = g(x)$ for all x

Surjective/Injective/bijective:

Surjective (onto):

f is surjective if every element of T actually appears as an output

$$\forall x \in T\ \exists x \in S \text{ such that } f(x)=y$$

Equivalently, range = codomain = $\mathrm{Im}(f) = T$.

Injective (one-to-one):

f is injective if different inputs gives different outputs.

$$\forall x_1, x_2 \in S, f(x_1) = f(x_2) \rightarrow x_1 = x_2$$

or contrapositive $x_1 \neq x_2 \rightarrow f(x_1) \neq f(x_2)$

bijective:

f is bijective if it is both injective and surjective

Inverse relation and inverse function:

For any relation $R \subseteq S \times T$, The inverse relation $R^{-1} \subseteq T \times S$ is

$$R^{-1} = \{(y,x) : (x,y) \in R\}$$

Always exists as a relation but it may not be a function

Let R be a relation from S to T

Then R^{-1} is a function from T to S if and only if R is bijective.

why?

if R is bijective:

- Injective $\rightarrow$ each y comes from at most one x. So in R^{-1} each y relates to at most one x. (uniqueness)
- Surjective $\rightarrow$ each $y \in T$ is hit by by some $x \rightarrow$ in R^{-1} each y has some x (Existence).

So R^{-1} is a function $T \rightarrow S$

when R is bijective and a function $f: S \rightarrow T$, we usually call R^{-1} the inverse function and write $f^{-1}: T \rightarrow S$

Example) $f(x) = x^3$, $\mathbb{R} \rightarrow \mathbb{R}$ is bijective; $f^{-1}(y) = \sqrt[3]{y}$

Identity Function:

For any set S, the identity function on S written id_S, is:

$$id_S = \{(x,x) : x \in S\}.$$

or as a rule: $id_S(x) = x$ for all $x \in S$.

* id_S is a function
* id_S is bijective

13.5. *Consider the relation* $R : \mathbb{R} \to \mathbb{R}$ *given by* $\{(x, y) : x^2 + y^3 = 1\}$. *Determine whether* R *is a well-defined function.* [Hint 209]

For every $x \in R$, there exists exactly one $y \in R$ such that $(x,y) \in R$

Existance: Fix an arbitary $x \in R$

Solve for y: $x^2 + y^3 = 1 \longrightarrow y^3 = 1 - x^2$, for any real number a the equation $y^3 = a$ has a real solution $y = \sqrt[3]{a}$, Here $a = 1 - x^2$, so for $y = \sqrt[3]{1-x} \in R$, for every $x \in R$, there exists exactly one $y \in R$ with $x^2 + y^3 = 1$

Uniqueness: Suppose for fixed x there are $y_1, y_2 \in R$ with

$$x^2 + y_1^3 = 1 \quad \text{and} \quad x^2 + y_2^3 = 1$$

Substract equations:

$$y_1^3 = 1 - x^2 = y_2^3$$

So $y_1^3 = y_2^3$, but the function $g(y) = y^3$ on R is injective, so equal cubes implies equal bases. So for each x there is at most one such y.

13.6. *Consider the relation* $R : \mathbb{R} \to \mathbb{R}$ *given by* $\{(x, y) : \sin^2 x + \cos^2 x = y\}$. *Determine whether* R *is a well-defined function.* [Hint 108]

Existance: for any real x, both $\sin x$ and $\cos x$ are real, so $\sin^2 x + \cos^2 x$ is a real number so for each we can define $y := \sin^2 x + \cos^2 x$ That give us some real y with $(x,y) \in R$.

Uniqueness: $\sin^2 x + \cos^2 x = 1$ for all $x \in R$ so the defining equation for R is really $y = 1$

13.7. *Consider the relation* $R : \mathbb{R} \to \mathbb{R}$ *given by* $\{(x, y) : y = \tan x\}$. *Determine whether* R *is a well-defined function.* [Hint 131]

Uniqueness: if $y_1 = \tan x$ and $y_2 = \tan x$ then $y_1 = y_2$

Existence: $\tan x = \dfrac{\sin x}{\cos x}$, when $\cos x = 0$, it is undefined i.e at $x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$

Counter Example: $x = \frac{\pi}{2}$ then $\cos \frac{\pi}{2} = 0$, so there is no real number such y that $y = \tan \frac{\pi}{2}$

13.8. *Consider the relation* $R : \mathbb{Q} \to \mathbb{Q}$ *given by* $\{(x, y) : xy = 1\}$. *Determine whether* R *is a well-defined function.* [Hint 127]

Uniqueness: if $xy=1$ and $xy'=1$ then $x(y-y')=0$ so if $x \neq 0$ then $y=y'$ $(x \neq 0)$

Existence: $y = \frac{1}{x}$ for any non zero rational x, $\frac{1}{x}$ is a rational number so fine for all $x \neq 0$

But domain is $\mathbb{Q}$ and $0 \in \mathbb{Q}$, for $x=0$ the solution becomes $0 \cdot y = 1$ which indeed has no solution $y \in \mathbb{Q}$

13.9. *Consider the relation* $f : \mathbb{R} \to \mathbb{R}$ *given by* $\{(x, y) : y = \sqrt{x}\}$. *Restrict the domain so that the resulting restricted relation becomes a function.* [Hint 192]

$$S' = [0, \infty) = \{x \in \mathbb{R} : x \geq 0\}$$

$$f' : S' \to \mathbb{R}, \quad f' = \{(x, y) \in S' \times \mathbb{R} : y = \sqrt{x}$$

13.12. *Consider the function* $f : \mathbb{Z} \to \mathbb{Z}$ *given by* $f(x) = x + 3$. *Prove that* f *is bijective.* [Hint 218]

we need to show it is injective and surjective

injective: Assume $f(a) = f(b)$ for some $a, b \in \mathbb{Z}$ Then

$$a+3 = f(a) = f(b) = b+3$$

subtract 3 from both side

$$a = b$$

Surjective: let $y \in \mathbb{Z}$ be arbitrary, we need to find an integer x such that

$$f(x) = y$$

take

$$x = y - 3$$

since $\mathbb{Z}$ is closed under subtraction, $x \in \mathbb{Z}$, now compute

$$f(x) = f(y-3) = (y-3)+3 = y$$

So for this x we have $f(x) = y$ because y was arbitrary, every integer y has some integer x with $f(x) = y$.

ch 13.3:

Identity function: for a set S the identity function is

$$id_S = \{(x,x) : x \in S\}$$

and in usual function notation: $id_S(x) = x$ for every $x \in S$.

- it does nothing to the element & just returns it
- As relation it is diagonal for all pairs.

why is id_S bijective?

injective: if $id_S(x) = id_S(x')$ then $x = x'$

surjective: for every $y \in S$ choose $x = y$, then $id_S(x) = y$

so identity function is always bijective

Compositions of functions: we have 3 sets and 2 functions:

$$F_1 : S_1 \to S_2, \quad F_2 = S_2 \to S_3$$

we define a new function:

$$F_1 \circ F_2 : S_1 \to S_3$$

by $(F_2 \circ F_1)(x) = F_2(F_1(x))$.

In relation language: $F_2 \circ F_1 = \{(x,z) : \exists y \in S_2, F_1(x) = y, F_2(y) = z\}$

why is $F_2 \circ F_1$ actully a function?

To be a function we need: for each $x \in S_1$ there is exaclty one $z \in S_3$ with $(x,z) \in F_2 \circ F_1$

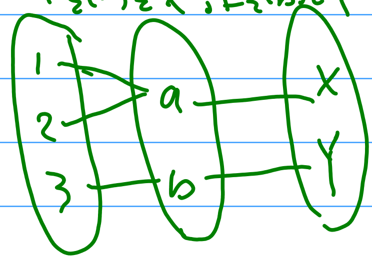
- Because F_1 is a function with domain S_1, for each $x \in S_1$ there is a unique $y \in S_2$ with $y = F_1(x)$
- Because F_2 is function with domain S_2, for that y there is a y is S_2 with $y = F_1(x)$
- So for each x there is exactly one z.

Ex) $S_1 = \{1,2,3\}$, $S_2 = \{a,b\}$, $S_3 = \{X,Y\}$, $F_1(1) = a$, $F_1(2) = a$, $F_1(3) = b$

$(F_2 \circ F_1)(1) = F_2(F_1(1)) = F_2(a) = X$ $\quad F_2(a) = X$, $F_2(b) = Y$

$(F_2 \circ F_1)(2) = F_2(F_1(2)) = X$

$(F_2 \circ F_1)(3) = F_2(b) = Y$



Composition & injective and surjective

$F_1 : S_1 \to S_2,\ F_2 : S_2 \to S_3$

(a) if F_1 and F_2 are injective then $F_2 \circ F_1$ is injective

$(F_2 \circ F_1)(x) = (F_2 \circ F_1)(x')$ — F_2 injective $\to F_1(x) = F_1(x')$

$F_2(F_1(x)) = F_2(F_1(x'))$ — F_1 injective $\to x = x'$

So $F_1 \circ F_2$ is injective

(b) if $F_2 \circ F_1$ is injective then F_1 is injective

$x, x' \in S_1$, with $F_1(x) = F_1(x')$

$y = F_1(x) = F_1(x')$ $\quad (F_2 \circ F_1)(x) = F_2(\ ) = (F_2 \circ F_1)(x')$

since $F_2 \circ F_1$ is injective then $x = x'$, so F_1 is injective

Inverse & identity:

Assume $F : S \to T$ is bijective and $F^{-1} : T \to S$ is inverse

1. $F^{-1} \circ F = id_S$
2. $F \circ F^{-1} = id_T$

Ex) let $F : \mathbb{R} \to \mathbb{R}$, $f(x) = x+3$, then $f^{-1}(x) = x-3$

- $F^{-1}(F(x)) = (x+3) - 3 = x$
- $F(F^{-1}(x)) = (x-3) + 3 = x$

Inverse of compositions:

Assume: $F_1 : S_1 \to S_2$ and $F_2 : S_2 \to S_3$ are bijective

Then $(F_2 \circ F_1)^{-1} = F_1^{-1} \circ F_2^{-1}$

Ex) let $F_1(x) = 2x\ (\mathbb{R} \to \mathbb{R})$

$F_2(x) = x + 1$

Then

$F_2 \circ F_1(x) = 2x + 1$ $\quad F_1^{-1}(x) = \frac{x}{2}$

inverse $\frac{x-1}{2}$ $\quad F_2^{-1}(x) = x - 1$

$(F_2 \circ F_1)^{-1}(x) = \frac{x-1}{2}$

Match

13.18. *Let* S, T *be sets, and* $f : S \to T$ *be a function. Prove that* $\mathrm{id}_T \circ f = f$. *[Hint 226]*

$\mathrm{id}_T(t) = t$ for all $t \in T$

$(g \circ f)(x) = g(f(x))$ for all $x \in S$

$(\mathrm{id}_T \circ f)(x) = \mathrm{id}_T(f(x))$

Domain of f is S Domain $\mathrm{id}_T \circ f$ is S

Let $x \in S$, Then $\mathrm{id}_T \circ f(x) = \mathrm{id}_T(f(x)) = f(x)$

13.19. *Let* S, T *be sets, and* $f : S \to T$ *be a function. Prove that* $f \circ \mathrm{id}_S = f$. *[Hint 280]*

$f(\underbrace{\mathrm{id}_S(x)}_{x}) = f(x)$

13.21. *Consider* F_1, F_2*, functions on* $\mathbb{R}$*, given by* $F_1(x) = e^x$*,* $F_2(x) = x^2$*. Recall that* F_1 *is injective (as proved in Example 13.11). Prove that* F_2 *is not injective, and that* $F_2 \circ F_1$ *is injective. This speaks to the "missing" part of Theorem 13.15.b.* *[Hint 197]*

$-1 \to 1$

$1 \to 1$

$2 \to 4$

$3 \to 9$

$F_2 \circ F_1 = F_2(F_1(x)) = F_2(e^x) = e^{2x}$

injective : $(F_2 \circ F_1)(x) = (F_2 \circ F_1)(y)$ then $x = y$

Let $x, y \in \mathbb{R}$ and $(F_2 \circ F_1)(x) = (F_2 \circ F_1)(y)$

$e^{2x} = e^{2y}$

we know F_1 is injective so $e^a = e^b \to a = b$

$2x = 2b \to x = y$